EXATHON 2026

DataTrust Ops

TRACK 4: THE WORK BEHIND THE WORK

Autonomous Data Reliability & Observability Platform

"Eliminating silent data decay before it poisons downstream business intelligence."

PROBLEM STATEMENT

DATA ACCURACY

Silent Corruption

Missing rows, duplicate IDs, and invalid field values go undetected during ingestion.

Impact: Corrupted datasets quietly flow into executive dashboards.

LATENCY & SLA

Temporal Drift

Pipelines halt or stall without throwing explicit fatal errors or crash logs.

Impact: Decisions are made on stale, hours-old business transactions.

OPEX OVERHEAD

Manual Triage Tax

Engineers waste critical sprint hours manually tracing log files and CSV anomalies.

Impact: High MTTR and severe loss of trust between teams.

! **The Core Problem:** *Traditional APM tools observe server uptime—not data integrity. Organizations lack unified, proactive data observability.*

PROPOSED SOLUTION

Key innovations

- Unified monitoring of pipelines, quality, freshness and incidents
- Independent data freshness evaluation
- Circuit-breakers stop high scores hiding severe problems
- Column-level validation with explainable incidents

Expected impact

- Earlier detection of data-quality issues
- Less manual inspection effort
- Better visibility into pipeline operations
- Greater confidence in business data
- More informed incident prioritisation

ARCHITECTURE & STACK



Frontend
React.js
Vite
Tailwind CSS
Recharts

Backend
Node.js
Express.js
REST APIs

Database
MongoDB
Mongoose

Data processing
Papa Parse
JavaScript validation and scoring

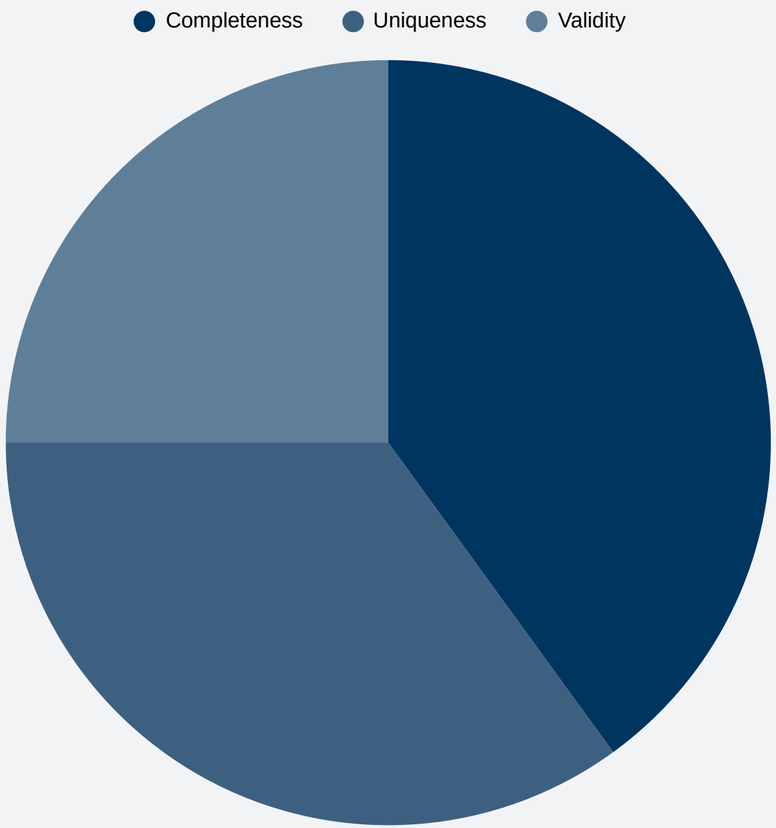
KEY MODULES

1	**Operations Dashboard**	Reliability scores, freshness, incidents and pipeline health
2	**Pipeline Monitoring**	Four simulated pipelines with status, logs and run controls
3	**CSV Quality Scanner**	Finds missing values, duplicates, invalid and empty data
4	**Reliability Scoring**	Weighted quality: completeness, uniqueness and validity
5	**Freshness Monitoring**	Checks data age against timestamp and SLA policies
6	**Incident Management**	Creates and tracks incidents for significant quality issues
7	**AI Incident Assistant**	Deterministic explanations and recommended next steps

IMPLEMENTED COMPONENTS

1. React-based responsive dashboard
2. Node.js and Express REST API backend
3. MongoDB database connectivity
4. CSV upload and automated validation
5. Data reliability scoring and operational circuit-breakers
6. Pipeline execution simulation
7. Incident creation and status management
8. Deterministic AI incident explanations

RELIABILITY SCORING



Weighted quality

Each dataset is scored on completeness (40%), uniqueness (35%) and validity (25%).

Operational circuit-breakers

Severe issues cap the grade, so a high weighted score cannot hide them.

Example: orders_missing_values.csv scores 94.7% but is graded C after circuit-breakers.

INNOVATION & IMPACT

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CONCLUSION & FUTURE SCOPE

DataTrust Ops transforms data reliability from reactive troubleshooting into proactive monitoring. By combining pipeline observability, automated quality validation, freshness assessment and incident management, it helps organisations maintain greater confidence in their data.

Future scope

- **Enterprise ETL platform integration**
- **LLM-powered root-cause analysis**
- **Real-time monitoring and alerts**
- **Predictive anomaly detection**
- **Cloud deployment and role-based access**

THANK YOU